



Retinal Review Workbench

Clinician User Manual

Clinician-led retinal review and dataset preparation

Software version	0.7.0
Manual version	1.2 (2026-09-23)
Document ID	RRW-CLIN-UM-001
Audience	Clinicians and reviewers using the review workstation
Owner	Retinal Review Workbench maintainers
Revision history	1.2 – image-selection pointer; documentation build path updated 1.1 – layout and screenshot refresh for the final v0.7.0 UI 1.0 – concise Quarto manual for the current clinician workflow
Screenshots	v0.7.0 UI with synthetic data only
Companion	Clinician Image Selection Guide (RRW-CLIN-SEL-001)

Research and clinical-support proof of concept for public or synthetic data. Model output is assistive evidence; human review remains authoritative.

Table of contents

1 Quick Start	2
2 Before You Begin	3
2.1 Intended users and scope	3
2.2 Where the images come from	3
2.3 Supported images	3
2.4 Model-assisted evidence	3
2.5 Workstation areas	3
3 Worklist and Confirm Image	4
3.1 Open the Worklist	4
3.2 Confirm Image	5
3.3 Admission exceptions	5
4 Review and Optional AI Assistance	6
4.1 Inspect the image	6
4.2 Read the model suggestions	6
4.3 Show or filter the overlays	7
5 Confirm DR Grade	7
5.1 Change a confirmed grade	8
6 Annotation Editor	9
6.1 Correct or remove an AI ROI	9
6.2 Draw human annotations	10
7 Working Through Cases	10
7.1 Reviewer default	10
7.2 Previous and Next	11
7.3 Autosave	11
7.4 Next action hint	12
8 Confirm Annotation	12
8.1 Change confirmed annotations	13
9 Dataset Readiness and Export	13
9.1 Check readiness	13
9.2 Export the manifest	14
10 Models & Audit and Traceability	14
10.1 Models & Audit	14
10.2 What the workstation records	15
10.3 Optional CVAT	15
11 Troubleshooting, Safety, and Glossary	16
11.1 Troubleshooting	16
11.2 Safety and privacy	16
11.3 Glossary	17

1 Quick Start

Retinal Review Workbench takes one admitted retinal image through a controlled review and turns the clinician's decisions into a traceable dataset record. Use this page as the everyday checklist; the following chapters explain each step.

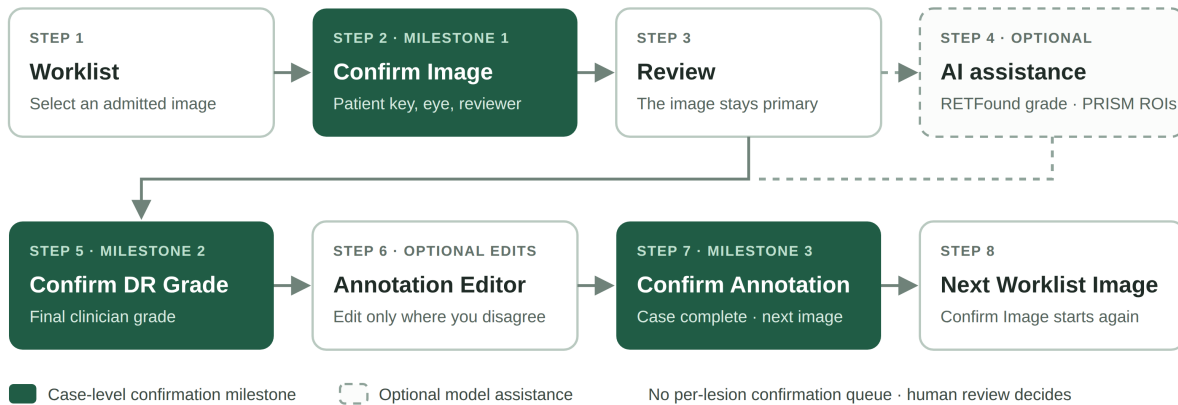


Figure 1.1: The clinician workflow. The three green boxes are the case-level confirmation milestones; AI assistance is optional.

The whole review is one forward line for one image at a time:

1. **Worklist** – choose **Confirm Image** in the case row.
2. **Confirm Image** – check the pseudonymous patient key, the eye, and your reviewer name, then choose **Confirm image & continue**. Review opens.
3. **Review** – inspect the retinal image and the optional RETFound grade and PRISM-DR lesion suggestions. Choose **Continue to clinician review**; nothing is decided on this page.
4. **Clinician Review** – select the **Final DR grade** and choose **Confirm DR Grade**. *Grading complete* appears and the Annotation Editor opens.
5. **Annotation Editor** – optional: click an AI ROI only where you disagree, change its class or box, and choose **Confirm** or **Remove**.
6. **Confirm Annotation** – finishes the image. *Case complete · Opening next image* appears and Confirm Image opens for the next unfinished Worklist image.

There is no separate “not confirmed” or senior-review button: a grade is simply *Not confirmed* until you choose **Confirm DR Grade**. Datasets (Section 9) shows DR-ready and Lesion-ready separately whenever you need them.

Important – three confirmations

Confirm Image, **Confirm DR Grade**, and **Confirm Annotation** are the only case-level confirmations. AI suggestions never need to be accepted or rejected one by one before the review can be completed.

2 Before You Begin

2.1 Intended users and scope

This manual is for clinicians and reviewers using the review workstation. Workstation installation, the hospital Model API, and backups are covered in the separate *Deployment and Operations Manual*.

Retinal Review Workbench is a clinician-controlled research and clinical-support workstation. It keeps source identity, patient and eye context, human decisions, annotations, model evidence, and export provenance together for review. It does not establish a diagnosis, disease truth, referral decision, or treatment decision. The final clinician review remains authoritative.

2.2 Where the images come from

You, or another authorized hospital user, choose each image from hospital-controlled storage, and the approved hospital process stages it into the Workspace input folder. The project team does not browse hospital storage or pre-select images. It monitors overall Refer/Not-refer, sex, and left/right coverage, so you do not manage dataset quotas yourself. What to choose, what to avoid, and what to do when unsure are covered in the one-page *Clinician Image Selection Guide* (docs/pdfs/CLINICIAN_IMAGE_SELECTION_GUIDE.pdf).

2.3 Supported images

The workstation accepts JPEG, PNG, TIFF, and supported single-frame ophthalmic fundus DICOM. MRI, OCT, PACS objects, non-fundus DICOM, unsupported SOP classes, multi-frame objects, and untested transfer syntaxes are outside model support. A recognized non-fundus DICOM is shown as **Unsupported modality**, not as a generic read failure.

2.4 Model-assisted evidence

- **RETFound** (retfound-aptos5) provides a five-class DR grade suggestion.
- **PRISM-DR** (prism-dr-5fold) provides optional lesion ROI suggestions.

A displayed confidence or score is a model score, not a calibrated clinical probability. An empty PRISM result does not prove that lesions are absent.

2.5 Workstation areas

Area	What you do there
Worklist	Find admitted cases, see their status, and start Confirm Image .
Review	Inspect the image and optional model assistance (read-only).
Clinician Review	Select and record the final DR grade.
Annotation Editor	Add human annotations or confirm, correct, or remove an AI ROI when needed.
Datasets	Check DR-ready and Lesion-ready status and export the manifest.
Models & Audit	Read model, inference, hash, and annotation provenance.
Settings	Select a Workspace, set the default reviewer, and manage hints.

Safety – human authority

Treat every model output as assistive evidence. Do not use a model score or an empty lesion result as a diagnosis, a referral decision, or proof that disease is absent.

3 Worklist and Confirm Image

3.1 Open the Worklist

Purpose. Start a review session in the approved Workspace.

Steps.

1. Start the workstation with the launcher provided by your operator and open `http://127.0.0.1:8000/app/`.
2. Check that the sidebar shows the correct **Active workspace**. If not, open **Settings** and select it.
3. In **Worklist**, choose **Scan input folder** when new images have been added.

Expected result. Each admitted image appears as one row with its patient and eye context, AI result, review status, and actions. Source files are never moved or changed.

Worklist				
Review admitted images, resolve patient and eye identity, and keep AI assistance separate from the clinician decision.				
Search filename or patient key		Filters	Sort: Filename	Scan input folder Refresh
Search by filename or pseudonymous patient key.				
4 cases - 4 need attention				
IMAGE	PATIENT / EYE	AI RESULT	REVIEW	ACTION
 img_01.jpg	Patient not linked Eye side needs confirmation Link patient and confirm eye	Grade 2 Suggestion available	Pending review	Confirm Image ...
 img_02.jpg	Patient not linked Eye side needs confirmation Link patient and confirm eye	Grade 2 Suggestion available	Pending review	Confirm Image ...
 img_03.jpg	Patient not linked Eye side needs confirmation Link patient and confirm eye	Not analyzed	Pending review	Confirm Image ...
 img_04.jpg	Patient not linked Eye side needs confirmation Link patient and confirm eye	Not analyzed	Pending review	Confirm Image ...

Figure 3.1: Worklist for the owner-provided demo workspace. Each row starts with **Confirm Image**; the Review column later shows **Grade confirmed** or **Review complete**.

3.2 Confirm Image

Purpose. Confirm the image context before analysis or clinical review. This is the first confirmation milestone.

Steps.

1. In the case row, choose **Confirm Image**. (After the first confirmation the row shows **Review**; **Confirm Image again** stays in the row menu)
2. Check the filename and the note above the fields.
3. Enter the **Pseudonymous patient key**, for example PAT0001. Leave it empty when the case should stay unlinked.
4. Set **Eye** to **Left**, **Right**, or **Unknown**.
5. Enter your **Reviewer name**. Select **Use as default reviewer on this workstation** to prefill it next time.
6. Choose **Confirm image & continue**.

Expected result. The patient and eye context, reviewer, and timestamp are saved and **Review** opens for this image.

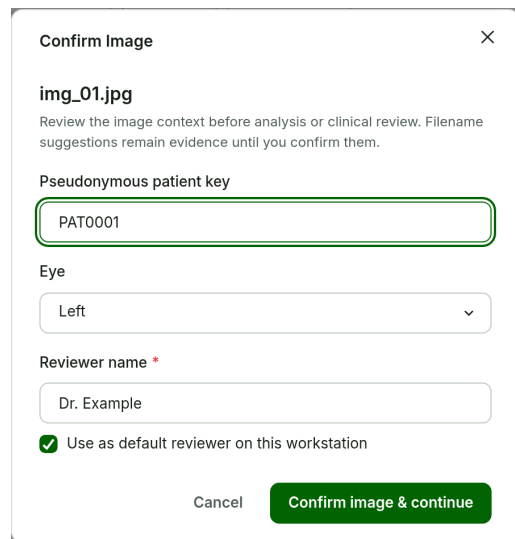


Figure 3.2: Confirm Image dialog.

Filename or resolver suggestions are evidence only; nothing is inferred from a filename until you confirm it.

3.3 Admission exceptions

Use the row status to decide the next action. **Resolve image readiness** and **Exclude from queue** are available in the same row menu. A recognized MRI or other non-fundus DICOM stays **Unsupported modality** and is never sent to RETFound or PRISM-DR.

Important – source files

Do not rename, move, overwrite, or recompress admitted source images. The workstation records the original source identity and works from it.

4 Review and Optional AI Assistance

4.1 Inspect the image

Purpose. Inspect the retinal image and any optional model evidence before the grade decision. No clinical decision is recorded on this page.

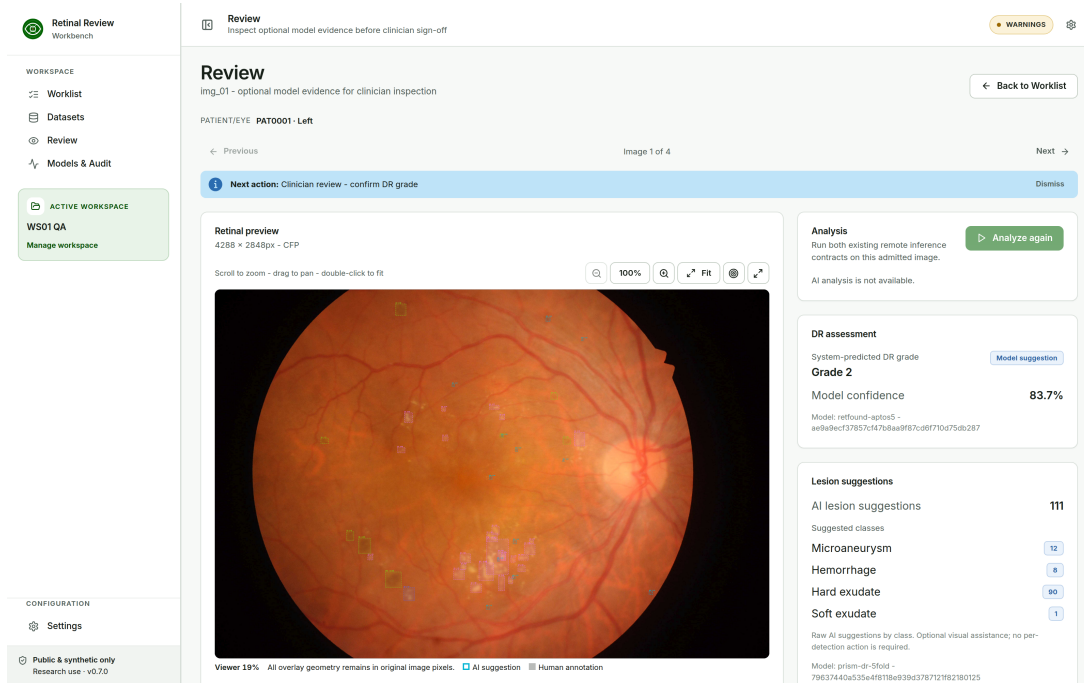


Figure 4.1: Review page: image on the left, optional model evidence and the single forward action on the right.

Steps.

1. Review opens after **Confirm Image**. You can also open it with **Review** in a confirmed Worklist row.
2. Use the viewer controls to zoom, fit, pan, inspect coordinates, or open the full-screen view.
3. Check the **Patient/Eye** line above the image.
4. Choose **Analyze** only when model assistance is wanted and the Model API is available. If it is not available, continue with manual review.
5. Choose **Continue to clinician review**. This is the only forward action on Review.

Expected result. The retinal image stays the main surface. The right column shows **Analysis**, **DR assessment**, and **Lesion suggestions**. Annotation work is not started from Review; it follows the grade decision (Section 6).

4.2 Read the model suggestions

- **DR assessment** shows the RETFound grade suggestion and its model confidence. This is a model score, not a calibrated probability.
- **Lesion suggestions** counts PRISM-DR suggestions by class. An untouched overlay is labelled with class and score, for example EX · 0.82.
- The lesion summary is not a task list. You do not need to inspect every AI ROI.

4.3 Show or filter the overlays

Use **Show AI suggestions** to show or hide the PRISM overlays, and the lesion filter to display **All lesion classes** or one class: **MA · Microaneurysm**, **HE · Hemorrhage**, **EX · Hard exudate**, or **SE · Soft exudate**. The counter shows how many raw AI suggestions are active.



Figure 4.2: Overlay controls below the viewer.

5 Confirm DR Grade

Purpose. Record the clinician's final DR grade. This is the second confirmation milestone.

Steps.

1. From **Review**, choose **Continue to clinician review**.
2. Check **Status: Not confirmed** and **Reviewer name**. The reviewer is prefilled when a workstation default is set.
3. Read the **AI suggestion** line, if a suggestion exists. It is never pre-selected.
4. Select the **Final DR grade** (Grade 0–4). **Confirm DR Grade** stays disabled until a grade is selected.
5. Add a **Remark (optional)** when useful.
6. Choose **Confirm DR Grade**.

Expected result. *DR grade confirmed · Grading complete* appears and the **Annotation Editor** opens for the same image, showing *DR grade confirmed · Grade X*. Keeping the model grade is recorded as **AI_ACCEPTED**, a different grade as **AI_CORRECTED**, and a grade without a model result as **MANUAL**. The human grade is the authoritative decision.

If you are not ready to decide, do not press **Confirm DR Grade**; the grade simply stays *Not confirmed*. There is no separate senior-review or not-confirm action. Historical senior-review records from earlier versions remain readable and are labelled *Legacy senior-review record*.

Clinician decision
Select the final DR grade, then confirm it.

Status: Not confirmed

Reviewer name *

Dr. Example

☒ Use as default reviewer on this workstation

AI suggestion: Grade 2 (optional evidence, not pre-selected)

Final DR grade

Grade 2

Remark (optional)

Add a review remark

Confirm DR Grade

This finalizes the DR grade and continues to annotation review.

DR grade confirmed
The confirmed grade is read-only. Reopen it only if the grade must change.

☒ **Grading complete**

Grade: Grade 2

Reviewer: Dr. Example

Confirmed at: 9/24/2026, 6:37:36 AM

Continue to Annotation Editor **Edit confirmed grade**

Figure 5.2: A confirmed grade is read-only.

Figure 5.1: Clinician decision before confirming.

5.1 Change a confirmed grade

Returning to a confirmed case shows **DR grade confirmed** with the grade, reviewer, and confirmation time. To change it, choose **Edit confirmed grade**. One warning appears for this edit session:

Edit confirmed DR grade?

This grade has already been confirmed.
Continuing will reopen the grading step and create a new review revision.

Cancel **Continue editing**

Figure 5.3: One warning per edit session.

Choose **Continue editing** to reopen the grade. The new confirmation creates a new review revision; the earlier decision stays in the review history.

6 Annotation Editor

Purpose. Record human annotations only where you disagree with the AI suggestions or want to add a finding. The Annotation Editor opens automatically after **Confirm DR Grade**.

AI suggestions are optional visual evidence. You only need to edit a region when you disagree or want to record a human annotation. Confirming annotations completes this case; it does not mean every AI ROI was individually verified. Untouched AI suggestions stay AI suggestions; they never become human labels by themselves.

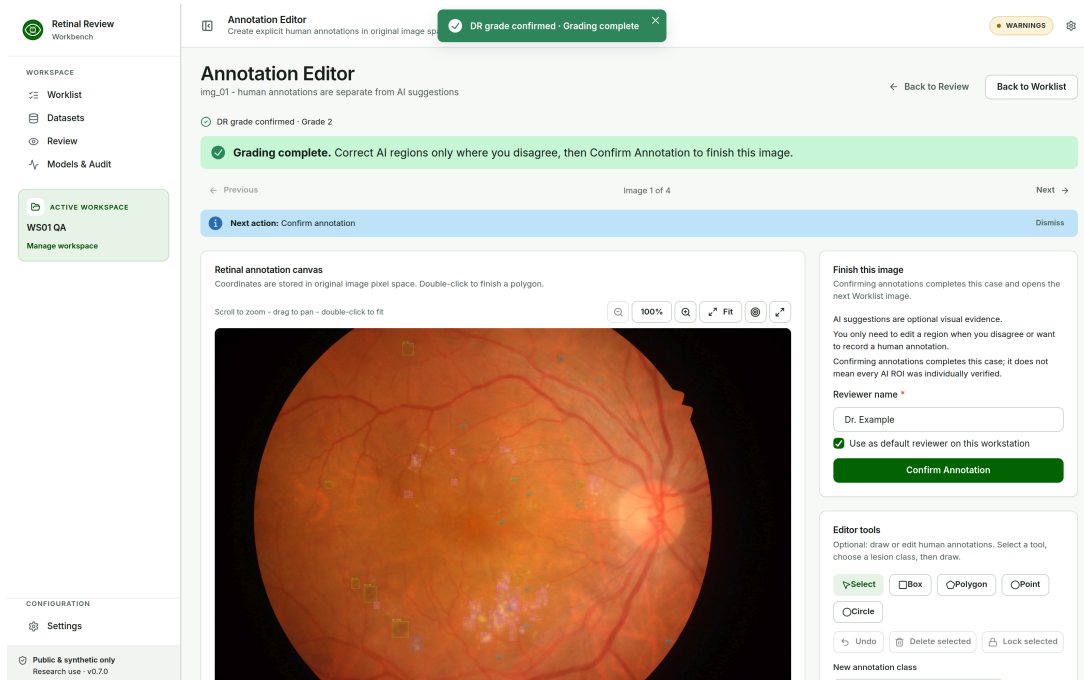


Figure 6.1: Annotation Editor after grade confirmation: canvas, **Finish this image** panel, and optional editor tools.

If an image is opened here before its grade is confirmed, a warning with **Open Clinician Review** appears and **Confirm Annotation** stays disabled. Your draft is kept.

6.1 Correct or remove an AI ROI

Steps.

1. Click an AI ROI on the image (inline or full-screen). It is highlighted, other ROIs dim, and the **AI suggestion** card opens beside it without covering it. Near the edge of the viewer the card moves to the other side or docks in a corner.
2. Read **Model score** and **Model: PRISM-DR**.
3. To change the class, choose it in **Lesion class**. The score line then reads *for AI class ...*: the score belongs to the model's class, not to your correction.
4. To change the box, drag it or drag a handle. The original AI box stays visible underneath.
5. Choose **Confirm** to record the ROI as a human annotation, **Remove** to remove it from the active reviewed result, or **Close** (or Escape) to leave without a change.

Expected result. *Annotation confirmed* or *AI suggestion removed* appears briefly and you stay on the image. A corrected ROI is shown as **clinician corrected** without a score. The original AI class,

score, geometry, and model identity stay in the review record; a removed suggestion stays in the audit record.

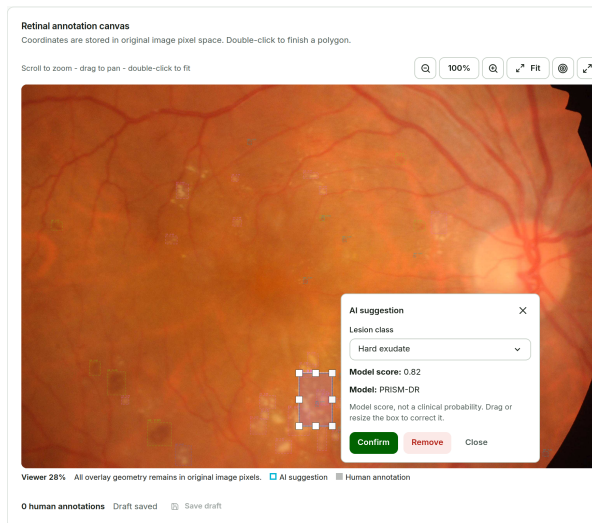


Figure 6.2: Selected AI ROI with its card beside it.

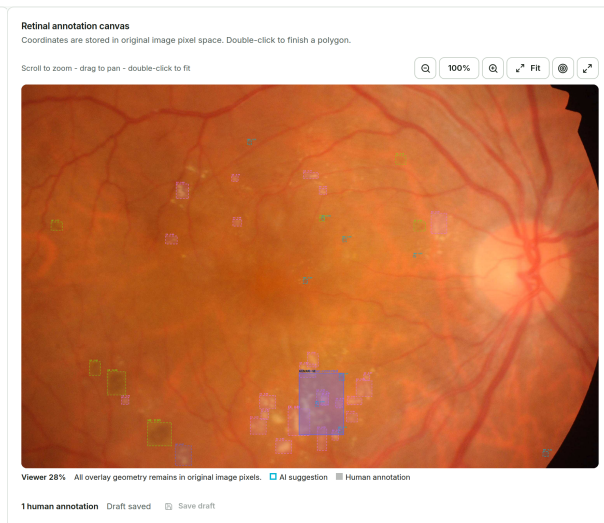


Figure 6.3: After correction: HUMAN annotation, no inherited score.

Important – score meaning

A model score belongs to the class and box the model predicted. If PRISM suggests Hemorrhage · 0.55 and you change the class to Hard exudate, the result is **Hard exudate – clinician corrected**. It is never shown as Hard exudate · 0.55.

6.2 Draw human annotations

Under **Editor tools**, choose **Box**, **Polygon**, **Point**, or **Circle** and a **New annotation class**, then draw. Double-click to finish a polygon. **Select**, **Undo**, **Delete selected**, and **Lock selected** adjust your own annotations. Human annotations are stored in original image pixels and remain separate from the AI evidence. Dense geometry work can use the optional CVAT round trip (Section 10.3).

7 Working Through Cases

These conveniences reduce repetitive work. None of them confirms a milestone or changes clinical state on its own.

7.1 Reviewer default

Select **Use as default reviewer on this workstation** in any reviewer field, or set **Default reviewer** in **Settings**, to prefill new or unattributed review forms.

Review preferences
Set a workstation default for new review records. It is editable and is not an authenticated identity.

Default reviewer

Dr. Example

New or unattributed review forms use this value. Clear it for a blank form; previously saved reviewers are unchanged.

☒ Show next action hints

Hints are guidance only. They do not change clinical state and are stored on this workstation.

Figure 7.1: Review preferences in Settings.

The value is stored only in this browser. It stays editable, it is not authentication, and it never rewrites reviewer names already recorded. On a shared workstation, change or clear it when the user changes.

7.2 Previous and Next

Previous and **Next** move through the Worklist order shown in the middle (for example Image 1 of 4) and open the other image at Review. Moving never saves a confirmation. While the current image is not complete, or has unsaved edits, one warning appears for that switch:

Switch to another image?

You are still working on this image.

Switching will open a different Worklist case.

Your autosaved draft will be kept, but this case is not complete.

Stay on this image Switch image

Figure 7.2: Switching away from an unfinished image.

Choose **Stay on this image** to continue, or **Switch image** to open the other image; the autosaved draft is kept. **Back to Worklist** on Clinician Review and Annotation Editor asks the same question. A complete image switches without a warning.

7.3 Autosave

In the Annotation Editor, edits are saved in the background about one second after you stop editing, provided a reviewer name is entered. The status below the canvas shows **Saving...**, **Draft saved**, or **Save failed**. The small **Save draft** button saves immediately if you need it.

0 human annotations Draft saved Save draft

Figure 7.3: Draft status below the canvas.

A draft is not a confirmation; only **Confirm Annotation** completes the image.

7.4 Next action hint

The blue **Next action** line suggests the next step, for example **Confirm image**, **Clinician review - confirm DR grade**, **Confirm annotation**, or **Review complete**. It is guidance only. Choose **Dismiss** to hide hints on this workstation; turn them back on with **Show next action hints** in **Settings**.

8 Confirm Annotation

Purpose. Finish this image. This is the third confirmation milestone.

Steps.

1. Make any optional ROI changes. You do not need to inspect every AI ROI.
2. Check **Reviewer name** in **Finish this image**.
3. Choose **Confirm Annotation**.

Expected result. Any pending draft is saved, then the reviewer, timestamp, and a hash of the active annotation set are recorded. *Case complete · Opening next image* appears, the Worklist shows **Review complete** for this image, and **Confirm Image** opens for the next unfinished Worklist image. When every image is complete, the Worklist says so.

Finish this image

Confirming annotations completes this case and opens the next Worklist image.

AI suggestions are optional visual evidence.

You only need to edit a region when you disagree or want to record a human annotation.

Confirming annotations completes this case; it does not mean every AI ROI was individually verified.

Reviewer name *

Dr. Example

☒ Use as default reviewer on this workstation

Confirm Annotation

Figure 8.1: Finish this image.

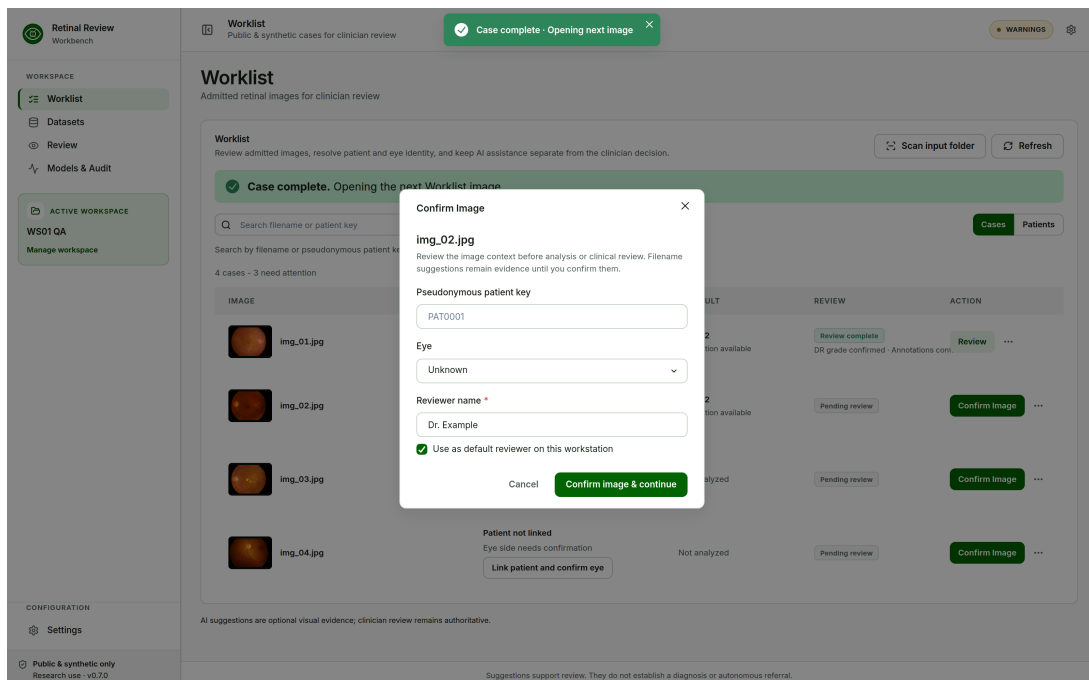


Figure 8.2: Case complete: the next Worklist image opens at Confirm Image.

8.1 Change confirmed annotations

A confirmed image shows **Annotations confirmed** with the reviewer and time, and the editor is read-only. Choose **Edit confirmed annotations** (or click an AI ROI) to reopen it; one warning appears per edit session. Any later edit changes the hash, so the image needs **Confirm Annotation** again before it is Lesion-ready.

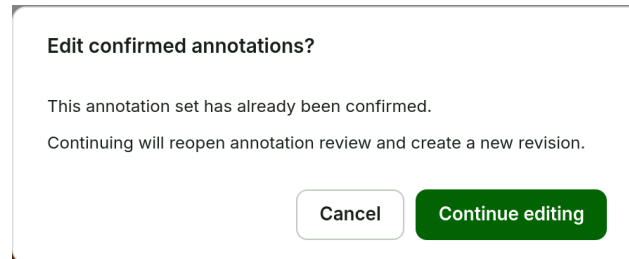


Figure 8.3: One warning before reopening confirmed annotations.

Confirming an empty set is allowed; it records that the set was reviewed, not that lesions are absent.

9 Dataset Readiness and Export

9.1 Check readiness

Purpose. See which cases are ready for each task. Classification readiness and lesion readiness are separate states.

Steps.

1. Open **Datasets**. The summary line shows **Images**, **Annotations**, **DR-ready**, and **Lesion-ready** for the active Workspace.
2. Choose a tab: **All**, **DR-ready**, **Lesion-ready**, **Needs review**, or **Excluded**.
3. Read the two status chips in **Dataset status**: one for DR, one for lesions.
4. Fix a case in **Clinician Review** or **Annotation Editor** when its status asks for it.

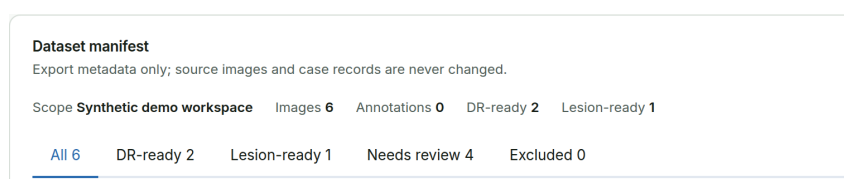


Figure 9.1: Dataset manifest summary with separate DR-ready and Lesion-ready counts.

PATIENT / EYE	CLINICIAN GRADE	ANNOTATIONS	DATASET STATUS
PAT0002 / Right	Grade 1	None	DR: Ready Lesions: Ready
PAT0003 / Left	Grade 0	None	DR: Ready Lesions: NO ANNOTATION CONFIRMATION

Figure 9.2: The DR-ready tab. PAT0002 is both DR-ready and Lesion-ready; PAT0003 is DR-ready but still needs annotation confirmation.

Expected result.

- **DR-ready** needs valid source provenance, an included fundus image, and a final clinician grade with reviewer and timestamp.
- **Lesion-ready** needs valid source provenance, an included fundus image, and a current **Confirm Annotation** whose hash matches the active set.

One task can be ready while the other is not. AI-only evidence is never promoted to ground truth by itself.

9.2 Export the manifest

Steps.

1. Check the readiness of the active Workspace.
2. Choose **Export manifest**.
3. Read the confirmation message.

✓ Export created: 6 images and 0 annotations.

Figure 9.3: Export confirmation.

Expected result. The package is written to the Workspace output folder: `manifest.json`, `images.csv`, and `annotations.csv` (schema `s4.dataset-manifest.v2`). Source images and case records are not changed. Untouched AI, human-authored, corrected, removed, and CVAT-imported states remain distinguishable in the export.

10 Models & Audit and Traceability

10.1 Models & Audit

Open **Models & Audit** when technical inspection is needed. It is read-only and is not another review queue. For the selected case it shows:

- model readiness, model IDs, revisions, runtime state, and limitations;
- the RETFound score context and the PRISM-DR lesion counts by class;
- **Case lineage & annotation provenance:** source SHA-256, analysis SHA-256, AI visual evidence, human annotations, CVAT imports, and inference metadata.

Case lineage & annotation provenance

Read-only evidence for this case. Hashes identify the source and exact analysis bytes; counts do not establish accuracy.

Source SHA-256

Analysis SHA-256

5a2d64fa5282f3774ad7c12da3ba81fedbec899514c9fdce25832f55250d6346

Not recorded

AI visual evidence	Human annotations	CVAT imported
1	1	0

Inference metadata

mock-global
 9/23/2026, 9:43:08 AM

synthetic-v1

mock-lesion
 9/23/2026, 9:43:08 AM

synthetic-v1

Figure 10.1: Case lineage for SYNTH_001. The synthetic fixture uses the explicit mock providers, so no analysis derivative is recorded.

10.2 What the workstation records

For every case the workstation keeps:

Record	Contents
Source identity	Original filename, source SHA-256, admission state; the source file is never modified.
Case context	Pseudonymous patient key, eye, reviewer, and timestamp from Confirm Image .
Model evidence	Model ID, revision, preprocessing, analysis SHA-256, grade suggestion, raw lesion output.
Clinician decisions	Final DR grade, review action (AI_ACCEPTED, AI_CORRECTED, MANUAL), remark, reviewer, time.
Annotations	Human annotations, AI ROI corrections and removals with the original AI evidence, and the confirmation hash.
Dataset state	DR-ready and Lesion-ready status and the exported manifest.

10.3 Optional CVAT

CVAT Online can be used for dense annotation or geometry correction when your operator has configured it. The round trip preserves source and prediction hashes. CVAT is optional and is not needed for offline review or export; the local workstation remains the source of the final case state.

11 Troubleshooting, Safety, and Glossary

11.1 Troubleshooting

What you see	What to do
Unsupported modality	The DICOM is recognized but is not supported fundus input. Do not send it to the models.
AI analysis is not available	Continue image review and human decisions. Ask the operator to check the Model API and the workstation connection.
No PRISM overlays	Check Show AI suggestions , the lesion filter, and whether a result exists. An empty result does not prove lesion absence.
Save failed	Stay on the page, check the reviewer name and connection, then choose Save draft again.
Case is not DR-ready	Check Confirm Image , the final grade, the reviewer, and the timestamp.
Case is not Lesion-ready	Check whether the annotation set changed after Confirm Annotation ; confirm it again.
Wrong reviewer prefilled	Edit the field, or change Default reviewer in Settings .
CVAT is unavailable	Continue local review. Retry once the operator confirms URL, project, credentials, and network.

11.2 Safety and privacy

Safety – clinical use

This is a research and clinical-support proof of concept. Model output is assistive evidence and human review remains authoritative. The product does not establish diagnosis, referral, treatment, or disease truth.

Safety – data handling

Use only approved public, synthetic, or authorized data. Never put PHI, private DICOM headers, tokens, passwords, CVAT credentials, model weights, or runtime databases in screenshots, examples, exports, or source control.

11.3 Glossary

Term	Meaning
AI suggestion	A model output shown to support inspection.
Confirmation milestone	One of Confirm Image , Confirm DR Grade , Confirm Annotation .
DR-ready	The case meets the dataset rules for the DR-grade task.
Lesion-ready	The case meets the dataset rules for the lesion-annotation task.
Model score	A value returned by a model; not a calibrated clinical probability.
Original AI evidence	Class, score, geometry, identity, model, and provenance returned by inference, kept unchanged after human edits.
Reviewer default	A browser-local name used to prefill new forms; not authentication.